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THE EGYPTIAN E-LEARNING UNIVERSITY

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Information Technology

IT438

Communication Technology

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Introduction to Communications

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Introduction

- Communication – Basic process of exchanging information from one location (source) to destination (receiving end).

- Refers to the process of sending, receiving and processing of information/signal/input from one point to another point.



Communication Systems Components

- Any communication system consists of three basic blocks:
  - Transmitter
  - Receiver
  - Communication Channel



- A transmitter prepares the data (information) to be transmitted in the appropriate format in order to be transmitted over the communication channel.

Communication System Components (cont.)

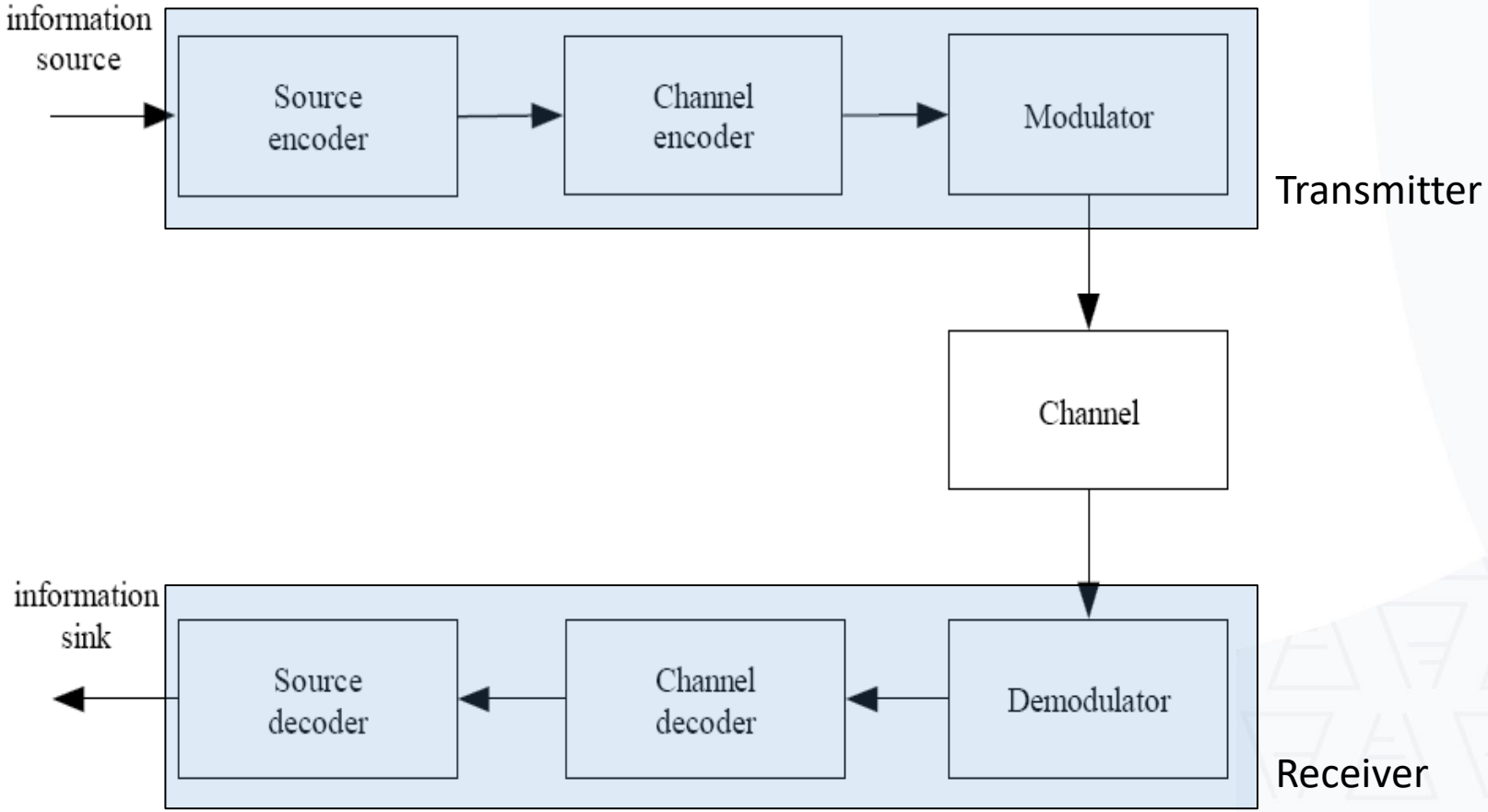


Figure 1.1: Block diagram of a communication system

Communication System Components (cont.)

1. Information Source

- To generate the baseband signal carrying the message information.
- Includes the Input Transducer (e.g., microphone for sound, sensor for data) that converts the physical phenomenon into a proportional electrical voltage or current.
- Digital Conversion (ADC): For digital systems, the Analog-to-Digital Converter performs the following critical steps:
  a) Sampling: Converts the continuous-time signal to a discrete-time signal.
  b) Quantization: Converts the continuous-amplitude samples into a finite set of discrete amplitude levels.
  c) Encoding: Maps the quantized levels to binary codes, producing the raw bit stream (e.g., Pulse Code Modulation - PCM).

Communication System Components (cont.)

Source encoder/decoder (Compression)

- The source encoder maps the signal produced by the source into a digital form.
- The mapping is done so as to remove redundancy in the output signal and also to represent the original signal as efficiency as possible (using as few bits as possible).
- The mapping must be such that an inverse operation (source decoding) can be easily done.
- If the message is a voice signal, the source encoder would convert the analog voice signal into a digital form (binary data) and compress it (e.g., MP3 compression for audio).
- Focus: Efficient representation of the message, removing unnecessary bits.

Communication System Components (cont.)

Channel encoder/decoder

- Maps the input digital signal into another digital signal in such a way that the noise will be minimized.
- When transmitting binary data, the channel encoder could add extra bits (redundancy) that help detect and correct errors.
- Focus on Error detection and correction to ensure reliable transmission over noisy channels.

Modulator

- Modulation provides for efficient transmission of the signal over channel.
- Most modulation schemes impress the information on either the amplitude, phase or frequency of a sinusoid.

Communication System Components (cont.)

Communication Channel

1- Guided Channels

A medium that provides a controlled path for signal propagation by confining the energy within or along its physical structure.

Types of Guided Channels

–Separately insulated
–Twisted together
–Often "bundled" into cables
–Usually installed in buliding when built



(a) Twisted Pair Cable

- Consists of two insulated copper wires twisted around each other.
- Twisting reduces electromagnetic interference (EMI) and crosstalk.
- Used in telephone lines, LANs (Ethernet), DSL.
- Bandwidth: a few MHz to several hundred MHz depending on shielding (UTP vs. STP).

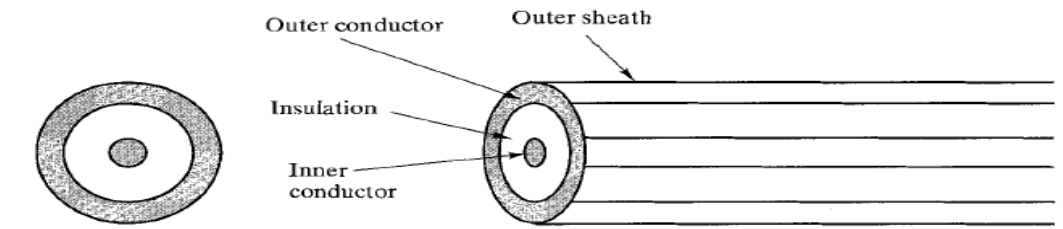
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Communication System Components (cont.)

Communication Channel (Cont.)

1- Guided Channels (Cont.)

(b) Coaxial Cable



–Outer conductor is braided shield
–Inner conductor is solid metal
–Separated by insulating material
–Covered by padding

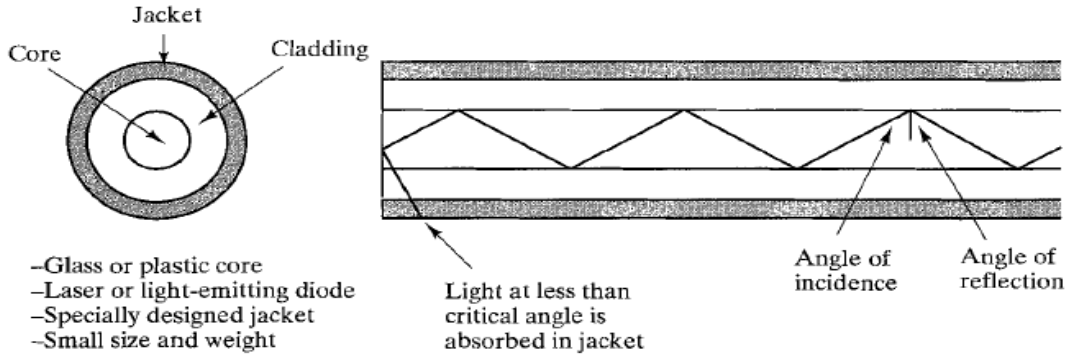
- Structure: central conductor, dielectric insulator, metallic shield, and outer cover.
- The shielding reduces external noise and allows higher bandwidth compared to twisted pair.
- Applications: cable TV, broadband internet, long-distance telephony.
- Frequency range: up to several GHz.

Communication System Components (cont.)

Communication Channel (Cont.)

1- Guided Channels (Cont.)

(c) Optical Fiber



- Uses total internal reflection to guide light signals inside a glass/plastic core.
- Very high bandwidth (terabits/s).
- Immune to EMI and secure against tapping.
- Types: **Single-mode fiber (SMF):** supports one propagation mode, used in long-distance, high-speed networks. **Multi-mode fiber (MMF):** supports multiple light paths, used in short-distance links.

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Communication System Components (cont.)

Communication Channel (Cont.)

1- Guided Channels (Cont.)

(d) Waveguides

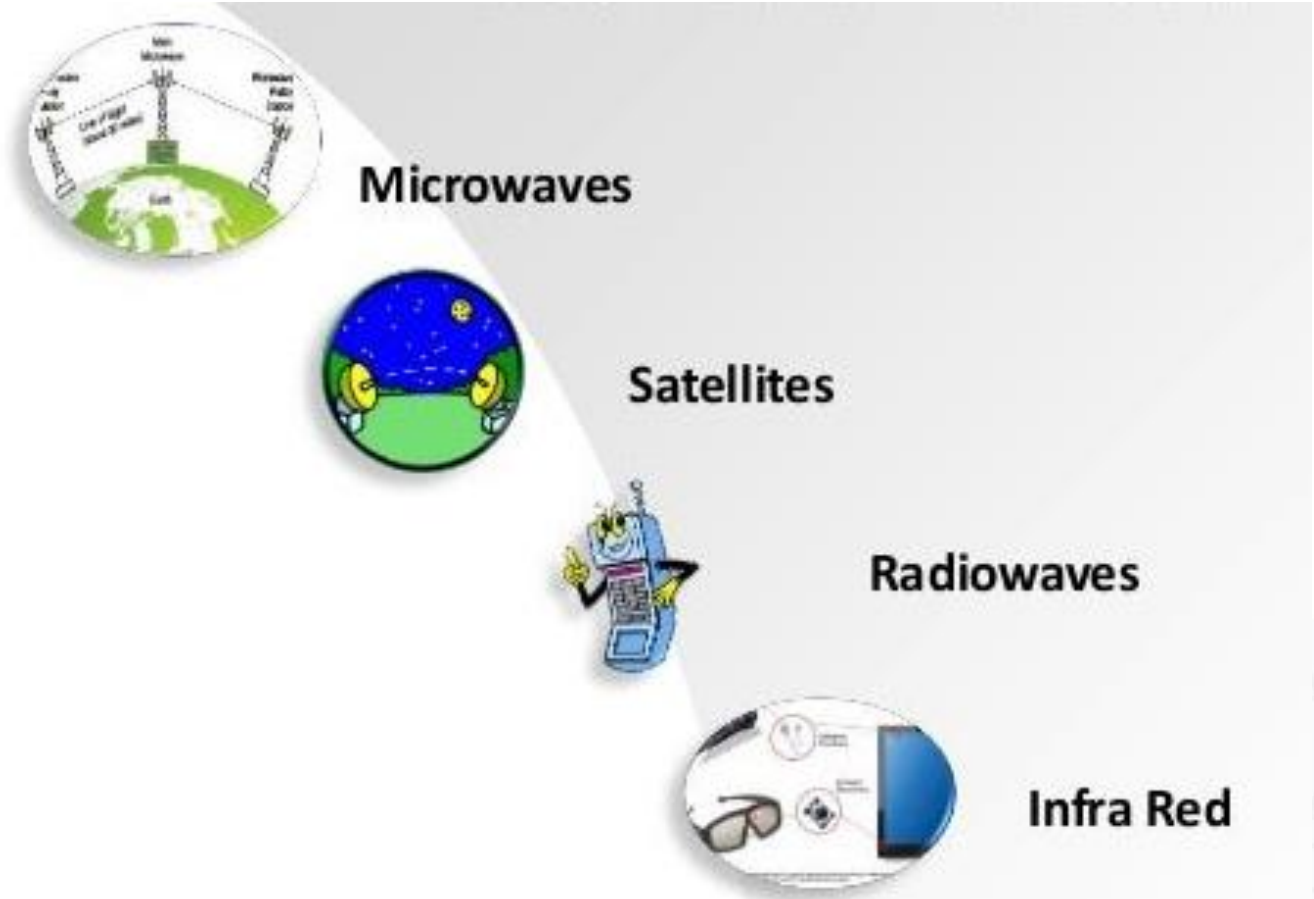
Hollow metallic or dielectric structures that guide high-frequency signals (microwave, millimeter-wave).

Signals are confined within boundaries through reflection.

Used in radar, satellite communication, and microwave links.



Examples of Unguided Comm. Channels



Why Different Types of Comm. Systems?

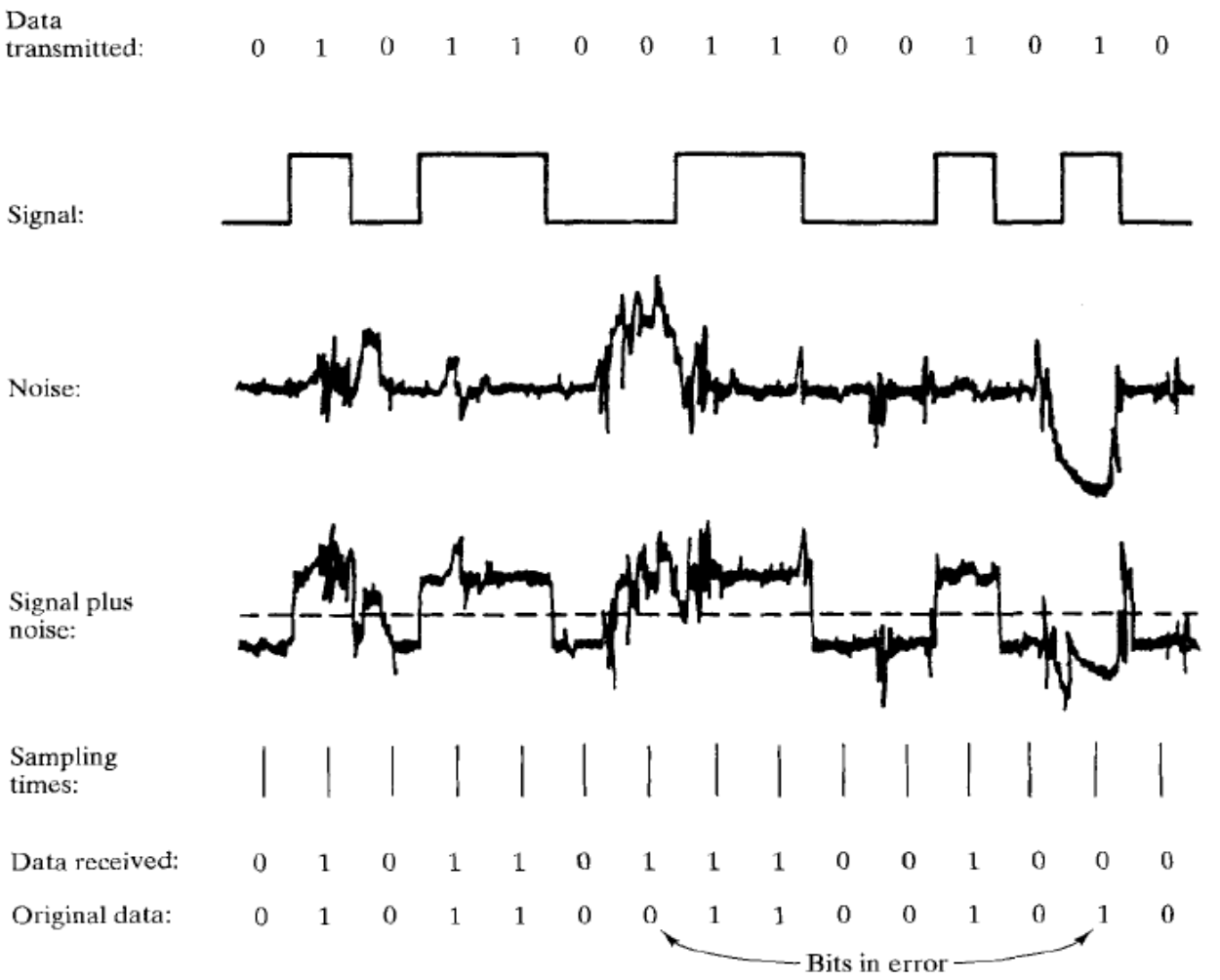
There are several factors that give rise to the need for different types of communication systems:

- The nature of the communication channel (undersea communication requires optical fiber cables)
- The nature of the application (mobile applications needs wireless systems)
- Required level of quality (performance and quality of the received signal)
- Cost

Problems that Face Comm. Systems

- Any communication system can be subject to three main sources of problems:
  - **Noise:** undesired effect from the communication environment. Usually, it is not under our control (your system has to deal with it).
  - **Interference:** due to superposition of two or more signals. May result from bad design of communication systems (interference of voice channels in telephone systems, for example)
  - **Jamming:** intentional interference that aims at destroying the quality of the transmitted signal to prevent transmission.

Problems that Face Comm. Systems (cont.)



Problems that Face Comm. Systems (cont.)

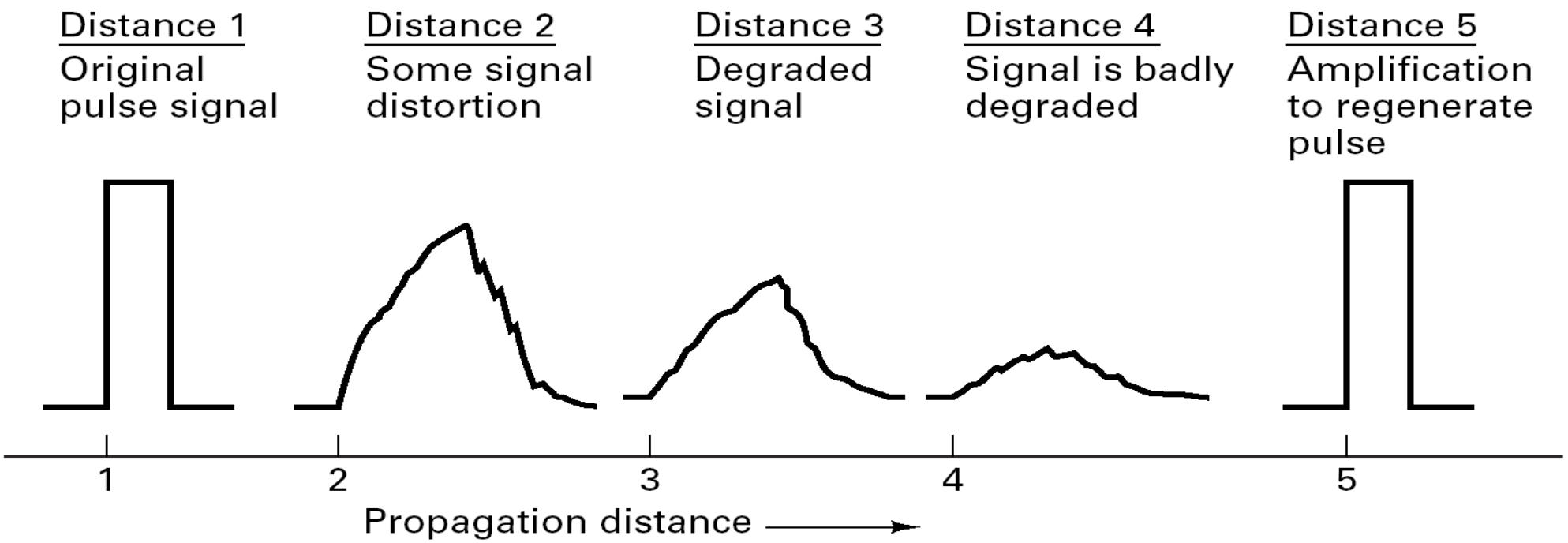


Figure 1.1 Pulse degradation and regeneration.

Data versus Signal

- Differentiate between two terms: data (information) and signal.
- Data (information): are generated by the application and need to be transmitted to some receiver.
- Signal: is the representation of the data in the communication system.
- Data are generated from the source (application) and signals are generated from the transmitter of the communication system we build.
- Speaking in a microphone: data (information) is what I say, whereas signal is what moves inside the wire of the microphone (electricity).

Types of Data and Signal

- Depending on the application nature, we have four possible combinations of data and signals:
  - Analog Data, Analog Signal: Traditional telephone system.
  - Analog Data, Digital Signal: Voice over IP (VoIP), digital music streaming.
  - Digital Data, Analog Signal: Modems (used in early internet connections), Optical networks.
  - Digital Data, Digital Signal: Ethernet, Wi-Fi, modern computer networks.
- Since any communication system deals with signals, we first need to understand the nature and types of signals.



THANK YOU

QUESTIONS?





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